

GENETIC CODE & INHIBITION OF PROTEIN SYNTHESIS

GENETIC CODE:

- The information needed to direct the synthesis of protein is contained in the mRNA in the form of a genetic code.
- Codons are a group of three adjacent bases that specify the amino acids of protein.

CHARACTERISTICS OF GENETIC CODE:

- **Number of codons:** There are 64 possible codon sequences. Because four nucleotide bases A, G, C and U are used to produce the three base codons, there are therefore 4^3 or 64 possible codon sequences (Figure 20.9).
- **Stop or termination or nonsense codons:** Three of the 64 possible nucleotide triplets, UAA, UAG and UGA do not code for any amino acids, they are called nonsense codons that normally signal termination of polypeptide chains. These **nonsense** codons are arbitrarily named **amber**, **ochre** and **opal**.
- **The code is degenerate but unambiguous:** As there are 61 codons for 20 amino acids, one amino acid has more than one codon and the code is referred to as **degenerate**, indicating that there are redundancies. Although an amino acid may have more than one codon, each codon specifies only one amino acid. Thus, the genetic code is unambiguous. Degeneracy minimizes the deleterious effects of mutations.
- Codons that designate the same amino acid are called **synonyms**. Two amino acids methionine (AUG) and tryptophan (UGC) each have only one codon. The remaining amino acids have multiple codons, e.g. arginine is specified by six different codons (Figure 20.9).
- **The code is almost universal:** That is, the meaning of each codon is the same in almost all known organisms. Exceptions to the universality of the genetic code are found in human mitochondria, where the code:
 - UGA codes for tryptophan instead of serving as a stop codon
 - AUA codes for methionine instead of isoleucine
 - CUA codes for threonine instead of leucine.

- **The code is non-overlapping and without punctuation:** During translation, the code is read sequentially, without spacer bases, from a fixed starting point, as a continuous sequence of bases, taken 3 at a time,

Ex: A U G C U A G A C U U U is read as: AUG / CUA / GAC / UUU without “punctuation” between the codons.

WOBBLE HYPOTHESIS FOR CODON-ANTICODON INTERACTIONS:

- The genetic code assumes that each codon base pairs in antiparallel fashion with the anticodon of the tRNAs that are specific for the amino acid corresponding to the codons on mRNA.
- It was found, however, that a single tRNA could recognize several codons.
- For example, the anticodon IGC (some of the tRNAs contain nucleoside inosine) of yeast tRNA for alanine binds to three codons: GCU, GCC and GCA.
- The first two bases of these codons are the same, whereas the third is different, “**wobble.**”
- Wobble allows some tRNAs to recognize more than one codon.
- The non-standard base pairing occurs in the third position of the codon, the position that has the least effect on specifying a particular amino acid (Figure 20.9).

		Second letter				
		U	C	A	G	
First letter	U	UUU } Phe UUC } UUA } Leu UUG }	UCU } UCC } Ser UCA } UCG }	UAU } Tyr UAC } UAA* UAC*	UGU } Cys UGC } UGA* UGG Trp	U C A G
	C	CUU } CUC } Leu CUA } CUG }	CCU } CCC } Pro CCA } CCG }	CAU } His CAC } CAA } Gln CAG }	CGU } Arg CGC } CGA } CGG }	U C A G
	A	AUU } AUC } Ile AUA } AUG Met	ACU } ACC } Thr ACA } ACG }	AAU } Asn AAC } AAA } Lys AAG }	AGU } Ser AGC } AGA } Arg AGG }	U C A G
	G	GUU } GUC } Val GUA } GUG }	GCU } GCC } Ala GCA } GCG }	GAU } Asp GAC } GAA } Glu GAG }	GGU } GGC } Gly GGA } GGG }	U C A G

* Termination code

Figure 20.9: The genetic code

Crick has proposed a hypothesis to explain how a single tRNA can recognize several (degenerate) codons. Crick proposed four rules called **wobble hypothesis**. These four rules are as follows:

- 1) The first two bases of a codon pair in the standard way and have normal Watson Crick base pairing.
- 2) For a given amino acid codons that differ in either of the first two bases must be recognized by different tRNAs. For example, both UUA and CUA code for leucine, but are read by different tRNAs.
- 3) The first base of an anticodon determines whether a particular tRNA molecule reads more than one codon for a given amino acid:
 - I. When the first base of the anticodon is C or A, it can read only one codon.
 - II. When it is U or G, it can read two different codons.
 - III. When the wobble base of an anticodon is I (inosine), it can read three different codons.
- 4) It is not necessary to have 61 different types of tRNA to read all 61 possible code words. A minimum of 32 tRNAs is required to translate all 61 different codons for the amino acids.

INHIBITION OF PROTEIN SYNTHESIS:

Many antibiotics and toxins work by inhibiting protein biosynthesis. They act at different stages of protein biosynthesis. Antibiotics block protein biosynthesis in prokaryotes. Usually eukaryotic protein synthesis is not affected by these antibiotics.

- 1) **Puromycin:** It is derived from mold. It has structure very similar to that of 3' end of aminoacyl-tRNA. So, it get incorporated into growing polypeptide chain by replacing incoming aminoacyl tRNA. As a result peptidyl puromycin is formed. Further addition of amino acids to peptidyl puromycin is blocked. Hence it is discharged from ribosome and protein synthesis is terminated.
- 2) **Tetracyclins:** They inhibit protein synthesis by preventing the binding of aminoacyl tRNA to A site.
- 3) **Chloramphenicol:** It inhibits protein biosynthesis by blocking peptidyl transferase activity of 50S subunit.
- 4) **Erythromycin:** It binds to 50S subunit and inhibits translocation step.
- 5) **Streptomycin:** It binds to 30S subunit and cause misreading of genetic code. It also interferes with binding of initiating tRNA to ribosome.

- 6) **Tunicamycin:** It prevents attachments of oligosaccharide side chains to certain glyco proteins.
- 7) **Cyclo heximide:** It inhibits protein biosynthesis in eukaryotes by blocking peptidyl transferase activity of 60S eukaryotic ribosome.
- 8) **Diphtheria toxin:** It is produced by bacteria that cause diphtheria in children. It inhibits protein synthesis in eukaryotes. It inactivates elongation factor of eukaryotes.

Table 20.4: Inhibitors of protein synthesis

Antibiotic	Action
Streptomycin	It binds to the 30S subunit of prokaryotes at the A site, thereby inhibits chain elongation by preventing the binding of additional aminoacyl tRNA
Tetracycline	Binds to the 30S subunit and inhibits binding of aminoacyl tRNA to mRNA in prokaryotes
Chloramphenicol	Binds to the 50S ribosomal subunit and blocks the peptidyl transferase reaction in prokaryotes
Erythromycin	Binds to the 50S ribosomal subunit that inhibits the translocation reaction in prokaryotes
Lincomycin and Clindamycin	Binds to the 50S subunit and inhibits peptidyl transferase, thereby, preventing peptide bond formation in prokaryotes
Puromycin	Causes premature chain termination in both prokaryotes and eukaryotes
Cycloheximide	Inhibits peptidyl transferase activity of the 60S ribosomal subunit in eukaryotes



THANK YOU

CHECK OUT OUR WEBSITE FOR PHARM.D STUDY MATERIAL

www.Pharmdguru.com